Alnico Magnet Properties

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Alnico magnet properties were gleaned from the following sources:

Grade	Demag Curve	Conductivity
LNG10	IMI	Arnold (Cast Alnico 3) 1.7 MS/m
LNG12	IMI	Arnold (Cast Alnico 2) 1.5 MS/m
LNG37	IMI	Magnet Sales (Cast Alnico 5) 2.1 MS/m
LNG40	IMI (LNG37/LNG44 interpolation)	Magnet Sales (Cast Alnico 5) 2.1 MS/m
LNG44	IMI	Magnet Sales (Cast Alnico 5) 2.1 MS/m
LNG56	IMI	EEC (Cast Alnico 5-7) 2.1 MS/m
LNGT28	IMI	EEC (Cast Alnico 6) 2 MS/m
LNGT32	IMI	EEC (Cast Alnico 8) 1.9 MS/m
LNGT38	IMI	EEC (Cast Alnico 8) 1.9 MS/m
LNGT44	IMI	EEC (Cast Alnico 8) 1.9 MS/m
LNGT72	IMI	EEC (Cast Alnico 9) 1.9 MS/m
FLNG12	IMI	EEC (Sintered Alnico 2) 1.5 MS/m
FLNG34	IMI	EEC (Sintered Alnico 5) 2 MS/m
FLNGT38	IMI (same as “FLNG38”)	EEC (Sintered Alnico 8) 1.9 MS/m

Table 1: Typical ALNICO properties

To translate between various grade conventions, see this chart.
As a check, the IMI curves were compared with the typical demagnetization curves in Arnold's cast alnico brochure, which are drawn as vector artwork and so could be read exactly rather than traced. Where the grades correspond, the curves agree: LNG44 and Arnold's Alnico 5 are within 5 mT of each other out to 80% of the coercivity, and LNG12 and Alnico 2 within 30 mT. Most of the other IMI curves lie 25–70 mT below Arnold's at 20% of the coercivity, and further below near the knee. This is a difference of grade, not of reading: every IMI curve but one reproduces IMI's own tabulated remanence, coercivity and energy product for its grade to within about 6% (the exception, FLNG12, has an energy product 14% above its row, a row that is itself inconsistent, printing 1.4 kJ/m^3 beside 1.55 MGOe), and each sits at the nominal energy product of its IEC grade (LNGT72 at 72 kJ/m^3), while Arnold quotes one typical value per grade, usually the higher one (83 kJ/m^3 for Alnico 9). The IEC designations divide the old Alnico classes more finely; LNG37, LNG40 and LNG44 are all Alnico 5.
The pairing of LNG10 with Alnico 3 is the loosest in the table. The LNG10 curve agrees with IMI's own figures for the grade (0.60 T, 44 kA/m, 10 kJ/m^3), but LNG10 is not quite Alnico 3: Alnico 3 is the cobalt-free isotropic grade, with a higher remanence (0.70 T) and a lower coercivity (38 kA/m) at nearly the same energy product, and none of IMI's grades corresponds to it closely. The LNG10 entry is kept as the nearest available measured curve; its conductivity is Arnold's figure for Alnico 3.

References

International Magnaproducts, Cast Alnico, accessed 28 Jan 2023.
International Magnaproducts, Sintered Alnico, accessed 28 Jan 2023
Arnold Magnetics, Cast ALNICO Permanent Magnets, accessed 23 Mar 2026.

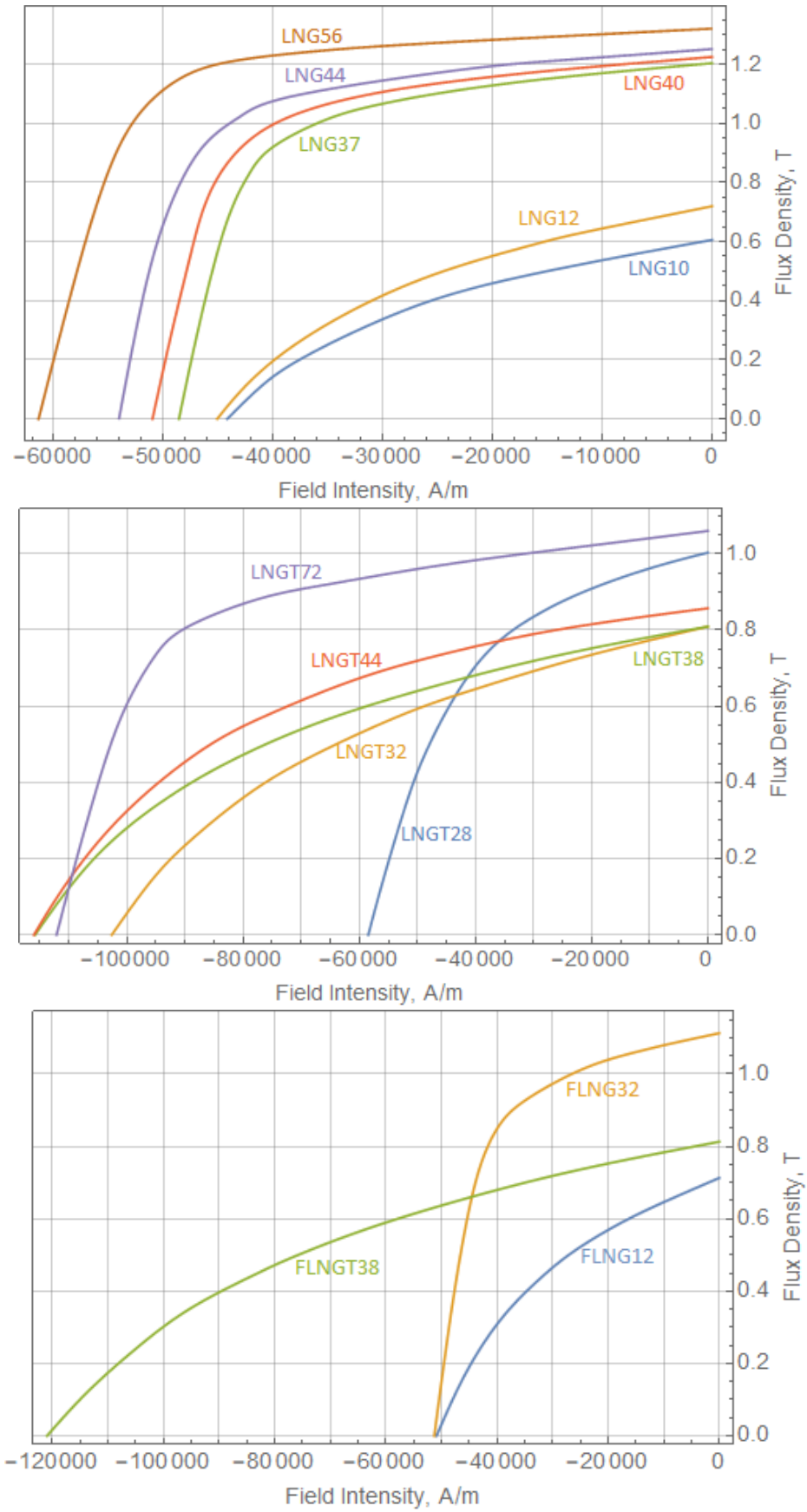


Figure 1: Plots of tabulated demagnetization curves

Magnet Sales, Properties of Alnico Magnets, accessed 25 Mar 2019.
Electron Energy Corporation, Alnico Magnets, accessed 29 Jun 2023.
Allstar Magnetics, Magnetic Properties of Cast Alnico, accessed 06 Aug 2025.
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